High School

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1 Logarithms

1.1 Exercise

Prove that if $x > 1$, then:
$2\log_2(x^2-1) - 3\log_2(x-1) = \log_2\frac{x^2+2x+1}{x-1}$

Solution

$D: x > 1$

$L = 2\log_2(x^2-1) - 3\log_2(x-1)$
$R = \log_2\frac{x^2+2x+1}{x-1}$

$L = \log_2(x^2-1)^2 - \log_2(x-1)^3 = \log_2\frac{(x^2-1)^2}{(x-1)^3} = \log_2\frac{\cancel{(x-1)^2}(x+1)^2}{\cancel{(x-1)^2}(x-1)} =$
$= \log_2\frac{x^2+2x+1}{x-1} = R \quad \blacksquare$

1.2 Exercise

Prove that $\log_{17}19 : \log_{18}19 = \log_{16}18 \cdot \log_{17}16$.

Solution

$\log_{17}19 \cdot \log_{19}18 = \log_{17}16 \cdot \log_{16}18$
$\log_{17}\cancel{19} \cdot \log_{\cancel{19}}18 = \log_{17}\cancel{16} \cdot \log_{\cancel{16}}18$
$\log_{17}18 = \log_{17}18$
$L = R \quad \blacksquare$

1.3 Exercise

Given the numbers $a = \log_3 2$ and $b = \log_{2^{2024}}3^{1012} + \log_{\frac{1}{3}}54$. Express b in terms of a.

Solution

$b = \frac{1}{2}\cdot\log_2 3 - \log_3(27\cdot 2) = \frac{1}{2}\cdot\frac{1}{a} - (\log_3 3^3 + a) = \frac{1}{2a} - 3 - a = \frac{1}{2a} - \frac{6a}{2a} - \frac{2a^2}{2a} =$
$= -\frac{2a^2+6a-1}{2a}$

1.4 Exercise

Given that $\frac{1}{\log_{p-q} 2} = 3$ and $(\log_{pq} 3)^{-1} = 2$. Prove that the number $\sqrt{p^2 - 1 + q^2}$ is rational.

Solution

$\log_2 p - q = 3 \quad \wedge \quad \log_3 pq = 2 \rightarrow pq = 3^2 = 9$
$p - q = 2^3 \ / \ ^2$
$(p-q)^2 = 64$
$p^2 + q^2 - 2pq = 64$
$p^2 + q^2 = 82$

$\sqrt{p^2 - 1 + q^2} = \sqrt{p^2 + q^2 - 1} = \sqrt{82 - 1} = \sqrt{81} = 9 \in \mathbb{Q} \quad \blacksquare$

1.5 Exercise

Given that $\log_a m = \sqrt{2}$. Evaluate $3 \log_{\sqrt{m}} am^2 - \log_{am} \frac{a^2}{m}$.

Solution

$3 \cdot 2 \log_m am^2 - \frac{\log_a \frac{a^2}{m}}{\log_a am} = 6 \cdot \log_m a + 12 - \frac{\log_a a^2 - \log_a m}{\log_a a + \log_a m} =$
$= \frac{6}{\sqrt{2}} + 12 - \frac{2-\sqrt{2}}{1+\sqrt{2}} = \frac{6\sqrt{2}}{2} + 12 - \frac{(2-\sqrt{2})(1-\sqrt{2})}{(1+\sqrt{2})(1-\sqrt{2})} =$
$= \frac{24+6\sqrt{2}}{2} - \frac{2-2\sqrt{2}-\sqrt{2}+2}{1-2} = \frac{24+6\sqrt{2}}{2} + 4 - 3\sqrt{2} = 16$

1.6 Exercise

Prove that for $a, b, c > 1$ and $n \in \mathbb{N}$ the number $\frac{\log_a b^3 \cdot \log_{b^2} c}{\log_{\sqrt[3]{a^n}} \sqrt[4]{c}}$ is even.

Solution

$\frac{\frac{3}{2} \log_a b \cdot \log_b c}{\frac{3}{4} n \log_a c} = \frac{\frac{3}{2} \cancel{\log_a c}}{\frac{3}{4} n \cancel{\log_a c}} = \frac{3}{2} \cdot \frac{4}{3} \cdot n = 2 \cdot n \quad \blacksquare$